

SCADA Alarm Troubleshooting Manual

Reference guide for the 29 alarm IDs used by the SCADA dashboard. The troubleshooting steps are intended as informational guidance for the dashboard/RAG prototype. Operators must follow the plant's approved operating, safety, switching, and maintenance procedures.

Generator System

Alarm ID: G1_CB

Problem: Generator 1 circuit breaker alarm.

Possible Causes: Breaker trip, protection operation, control circuit issue, or abnormal breaker status.

Recommended Troubleshooting:

1. Check the breaker indication and alarm details.
2. Check associated protection indications.
3. Check the breaker control/status circuit using the approved maintenance procedure.
4. If the fault remains, escalate to the electrical/maintenance team.

Alarm ID: G2_CB

Problem: Generator 2 circuit breaker alarm.

Possible Causes: Breaker trip, protection operation, control circuit issue, or abnormal breaker status.

Recommended Troubleshooting:

1. Check the breaker indication and alarm details.
2. Check associated protection indications.
3. Check the breaker control/status circuit using the approved maintenance procedure.
4. If the fault remains, escalate to the electrical/maintenance team.

Alarm ID: G3_CB

Problem: Generator 3 circuit breaker alarm.

Possible Causes: Breaker trip, protection operation, control circuit issue, or abnormal breaker status.

Recommended Troubleshooting:

1. Check the breaker indication and alarm details.
2. Check associated protection indications.
3. Check the breaker control/status circuit using the approved maintenance procedure.
4. If the fault remains, escalate to the electrical/maintenance team.

Alarm ID: TEMP_BEARING_G1

Problem: Generator 1 bearing temperature high.

Possible Causes: Bearing temperature may be elevated because of lubrication problems, cooling problems, excessive load, or sensor issues.

Recommended Troubleshooting:

1. Confirm the temperature trend and alarm status.
2. Check relevant lubrication and cooling indications.
3. Check for abnormal vibration or other related alarms.
4. Verify the temperature sensor/instrumentation according to the approved procedure.

Alarm ID: TEMP_BEARING_G2

Problem: Generator 2 bearing temperature high.

Possible Causes: Bearing temperature may be elevated because of lubrication problems, cooling problems, excessive load, or sensor issues.

Recommended Troubleshooting:

1. Confirm the temperature trend and alarm status.
2. Check relevant lubrication and cooling indications.
3. Check for abnormal vibration or other related alarms.
4. Verify the temperature sensor/instrumentation according to the approved procedure.

Alarm ID: TEMP_STATOR_G1

Problem: Generator 1 stator temperature high.

Possible Causes: Possible excessive electrical loading, cooling issue, or temperature instrumentation problem.

Recommended Troubleshooting:

1. Confirm the stator temperature trend.
2. Check generator load and cooling-system indications.
3. Check for related generator alarms.
4. Verify the temperature measurement system according to the approved procedure.

Transformer System

Alarm ID: TR1_CB

Problem: Transformer 1 circuit breaker alarm.

Possible Causes: Breaker trip, protection operation, control circuit issue, or abnormal breaker status.

Recommended Troubleshooting:

1. Check breaker status and the associated alarm.
2. Review transformer protection indications.
3. Check the breaker control/status circuit using the approved procedure.

4. Escalate persistent faults to the electrical/maintenance team.

Alarm ID: TR2_CB

Problem: Transformer 2 circuit breaker alarm.

Possible Causes: Breaker trip, protection operation, control circuit issue, or abnormal breaker status.

Recommended Troubleshooting:

1. Check breaker status and the associated alarm.
2. Review transformer protection indications.
3. Check the breaker control/status circuit using the approved procedure.
4. Escalate persistent faults to the electrical/maintenance team.

Alarm ID: TR3_CB

Problem: Transformer 3 circuit breaker alarm.

Possible Causes: Breaker trip, protection operation, control circuit issue, or abnormal breaker status.

Recommended Troubleshooting:

1. Check breaker status and the associated alarm.
2. Review transformer protection indications.
3. Check the breaker control/status circuit using the approved procedure.
4. Escalate persistent faults to the electrical/maintenance team.

Alarm ID: TEMP_TRANSFORMER_TR1

Problem: Transformer 1 temperature high.

Possible Causes: High load, cooling problem, abnormal transformer condition, or temperature sensor issue.

Recommended Troubleshooting:

1. Confirm the temperature trend and alarm.
2. Check transformer loading and cooling indications.
3. Review related transformer protection alarms.
4. Verify the temperature instrumentation according to the approved procedure.

Cooling System

Alarm ID: COOLING_PUMP_1

Problem: Cooling pump 1 alarm.

Possible Causes: Pump fault, motor/electrical supply issue, low flow, or control problem.

Recommended Troubleshooting:

1. Check pump running/fault indication.

2. Check motor and electrical supply indications.
3. Check available flow/pressure indications.
4. Inspect the pump through the approved maintenance procedure.

Alarm ID: COOLING_PUMP_2

Problem: Cooling pump 2 alarm.

Possible Causes: Pump fault, motor/electrical supply issue, low flow, or control problem.

Recommended Troubleshooting:

1. Check pump running/fault indication.
2. Check motor and electrical supply indications.
3. Check available flow/pressure indications.
4. Inspect the pump through the approved maintenance procedure.

Alarm ID: V_AUX_COOLING

Problem: Auxiliary cooling valve alarm.

Possible Causes: Valve position mismatch, actuator issue, control signal problem, or mechanical restriction.

Recommended Troubleshooting:

1. Check commanded versus actual valve position.
2. Check actuator status and related alarms.
3. Check the control signal/status.
4. Escalate persistent mechanical or actuator faults.

Alarm ID: TEMP_CONDENSER_EXHAUST

Problem: Condenser exhaust temperature high.

Possible Causes: Cooling performance issue, abnormal operating condition, or temperature sensor problem.

Recommended Troubleshooting:

1. Confirm the temperature trend.
2. Check cooling-water flow and related pump indications.
3. Review condenser/vacuum-related alarms.
4. Verify the temperature sensor if the reading appears abnormal.

Alarm ID: PRESS_VACUUM_TRIP

Problem: Condenser/vacuum pressure trip alarm.

Possible Causes: Vacuum deterioration, cooling problem, air ingress, or pressure instrumentation issue.

Recommended Troubleshooting:

1. Confirm the pressure/vacuum trend and related alarms.
2. Check condenser cooling indications.
3. Check for associated pump or valve alarms.
4. Follow the plant's approved trip/recovery procedure and escalate if abnormal conditions persist.

Steam & Boiler System

Alarm ID: V_MAIN_STEAM

Problem: Main steam valve alarm.

Possible Causes: Valve position mismatch, actuator/control problem, or abnormal steam-system condition.

Recommended Troubleshooting:

1. Check commanded and actual valve position.
2. Check actuator and control-system indications.
3. Review related steam/boiler alarms.
4. Follow the approved valve inspection/maintenance procedure.

Alarm ID: V_REHEAT_STEAM

Problem: Reheat steam valve alarm.

Possible Causes: Valve position mismatch, actuator problem, control signal issue, or abnormal steam condition.

Recommended Troubleshooting:

1. Check commanded and actual valve position.
2. Check actuator status and control signal.
3. Review related reheat-steam alarms.
4. Escalate persistent valve/actuator faults.

Alarm ID: PRESS_STEAM_DRUM_HI

Problem: Steam drum pressure high.

Possible Causes: Abnormal pressure rise, control problem, load/steam-flow mismatch, or pressure instrumentation issue.

Recommended Troubleshooting:

1. Confirm the pressure trend and alarm.
2. Check boiler/control-system indications.
3. Review related steam and safety alarms.
4. Follow the approved high-pressure operating procedure; do not bypass protective systems.

Alarm ID: V_DRAIN_VALVE

Problem: Drain valve alarm.

Possible Causes: Valve position mismatch, actuator fault, control signal issue, or mechanical restriction.

Recommended Troubleshooting:

1. Check commanded versus actual valve position.
2. Check actuator and control indications.
3. Review related drain/steam alarms.
4. Inspect the valve only under the approved maintenance procedure.

Lube Oil System

Alarm ID: LUBE_OIL_PUMP_1

Problem: Lube oil pump 1 alarm.

Possible Causes: Pump fault, motor/power problem, control issue, or abnormal oil-system condition.

Recommended Troubleshooting:

1. Check pump running/fault status.
2. Check motor and electrical supply indications.
3. Check lube-oil pressure/level indications.
4. Follow the approved pump inspection/recovery procedure.

Alarm ID: PRESS_LUBE_OIL_LOW

Problem: Lube oil pressure low.

Possible Causes: Low oil level, pump problem, filter restriction, leakage, or pressure-sensor/relay issue.

Recommended Troubleshooting:

1. Confirm the pressure trend and related alarms.
2. Check lube-oil level.
3. Check the operating pump and its status.
4. Check filter/flow indications and inspect for abnormal leakage using the approved procedure.
5. Verify the pressure sensor/relay if the reading appears inconsistent.

Protection System

Alarm ID: ESD_RELAY

Problem: Emergency shutdown relay alarm.

Possible Causes: Emergency shutdown/protection logic has operated or the relay/control circuit has a fault.

Recommended Troubleshooting:

1. Confirm the ESD indication and associated alarms.
2. Identify the initiating condition from the alarm/event log.
3. Do not bypass or defeat the protection system.
4. Follow the approved ESD investigation and recovery procedure.

Alarm ID: FIRE_SUPPRESSION_RELAY

Problem: Fire suppression relay alarm.

Possible Causes: Fire detection/suppression logic activation or relay/control fault.

Recommended Troubleshooting:

1. Confirm the fire alarm and affected zone/system.
2. Review associated detector and suppression indications.
3. Follow the site's fire-response procedure.
4. Do not reset or isolate protection equipment unless authorized.

Alarm ID: GRID_FREQ_RELAY

Problem: Grid frequency protection relay alarm.

Possible Causes: Grid frequency has moved outside a configured protection range or the relay/control system has reported a fault.

Recommended Troubleshooting:

1. Confirm grid-frequency trend and relay status.
2. Review related generator/grid protection alarms.
3. Record the event and follow the approved grid disturbance procedure.
4. Escalate persistent relay/instrumentation faults.

Alarm ID: OVERSPEED_TRIP_RELAY

Problem: Overspeed trip relay alarm.

Possible Causes: Overspeed protection has operated or the relay/speed-measurement system has a fault.

Recommended Troubleshooting:

1. Confirm speed indication and trip event.
2. Review related turbine/generator alarms.
3. Do not bypass or reset overspeed protection without authorization.
4. Follow the approved turbine trip investigation procedure.

Switchyard System

Alarm ID: BUS_BAR_2_CB

Problem: Bus bar 2 circuit breaker alarm.

Possible Causes: Breaker trip, protection operation, control circuit issue, or abnormal breaker status.

Recommended Troubleshooting:

1. Check breaker status and alarm details.
2. Review busbar and protection indications.
3. Check control/status circuits using the approved procedure.
4. Escalate persistent electrical faults.

Alarm ID: BUS_TIE_CB

Problem: Bus tie circuit breaker alarm.

Possible Causes: Breaker trip, protection operation, interlock/control issue, or abnormal breaker status.

Recommended Troubleshooting:

1. Check breaker status and related interlock/protection indications.
2. Review busbar conditions and associated alarms.
3. Check control/status indications using the approved procedure.
4. Follow the approved switching procedure; do not defeat interlocks.

Alarm ID: GRID_FEED_CB1

Problem: Grid feeder circuit breaker 1 alarm.

Possible Causes: Breaker trip, grid protection operation, control circuit issue, or abnormal breaker status.

Recommended Troubleshooting:

1. Check feeder breaker status.
2. Review grid and protection relay indications.
3. Check related control/status indications.
4. Follow the approved switching and restoration procedure.

Alarm ID: GRID_FEED_CB2

Problem: Grid feeder circuit breaker 2 alarm.

Possible Causes: Breaker trip, grid protection operation, control circuit issue, or abnormal breaker status.

Recommended Troubleshooting:

1. Check feeder breaker status.

2. Review grid and protection relay indications.
3. Check related control/status indications.
4. Follow the approved switching and restoration procedure.